



6_MCMC

6. Markov Chain Monte Carlo (MCMC)

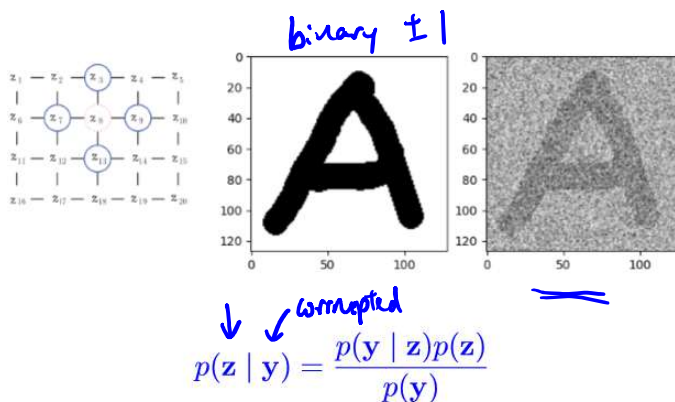
- 1 Stationarity
- 2 Metropolis-Hastings
- 3 Gibbs Sampling
- 4 Conclusion
- 5 Some Practice Problems

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Motivation



- For good performance, need a “reasonable” prior $p(\mathbf{z})$ that models relationship between neighboring pixels.
- But this makes $p(\mathbf{z} | \mathbf{y})$ a **high dimensional** distribution, not possible to have analytical form.
- We want to sample from $p(\mathbf{z} | \mathbf{y})$!

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$k q(\cdot) \approx \tilde{p}(\cdot)$ *unnormalized*

$\tilde{F}^{-1}(u)$, transformation

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- These can be used for low dimensional pdfs (e.g., scalar valued random variables) and as subroutines in other sampling methods. $\mathbf{z} \in \mathbb{R}^2$
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- Relies on the concept of “**stationary distribution**” of a Markov chain.
 - ▶ Want to sample from a target pdf $\pi(\mathbf{x})$.
 - ▶ Design a MC $Z_0, Z_1, \dots$ such that distribution of Z_n approaches $\pi(\cdot)$ as $n \rightarrow \infty$.

$Z_0 \rightarrow Z_1 \rightarrow Z_2 \rightarrow Z_3 \rightarrow \dots$

$\dots \rightarrow Z_n \rightarrow \dots$

$\approx$
stationary dist.
 $\equiv$ target pdf.

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Stationarity

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Stationary Distribution

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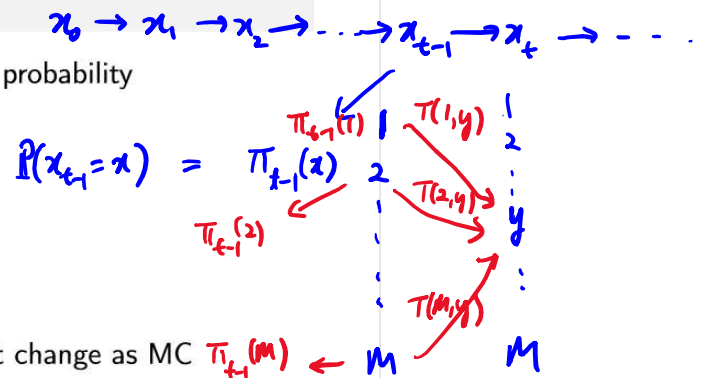
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- Also called invariant distribution - distribution does not change as MC proceeds.
- If there are M states,

$$\mathbf{T} = \begin{bmatrix} \mathbf{T}(1, 1) & \cdots & \mathbf{T}(1, M) \\ \vdots & & \vdots \\ \mathbf{T}(M, 1) & \cdots & \mathbf{T}(M, M) \end{bmatrix}$$



$$\begin{aligned} \pi_t(y) = P(x_t = y) &= \sum_{x=1}^M \pi_{t-1}(x) \mathbf{T}(x, y) \\ &\stackrel{||}{=} \sum_{x=1}^M \pi(x) \mathbf{T}(x, y) \end{aligned}$$

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- Stationary distribution satisfies $\pi \mathbf{T} = \pi$.

$$\pi = [\pi(1) \ \pi(2) \ \cdots \ \pi(M)]$$

$$\pi \mathbf{T} = \pi$$

$$\pi (\mathbf{T} - \mathbf{I}) = 0 \quad \text{linear system}$$

Conditions for Stationarity (Supplementary)

- Conditions for existence of stationary distribution:

Conditions for Stationarity (Supplementary)

exist?

$$\downarrow$$

$$\pi \mathbf{T} = \pi$$

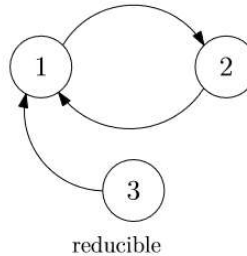
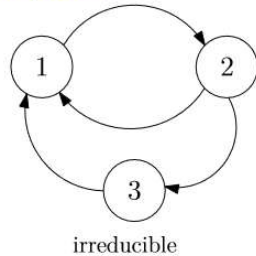
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↓
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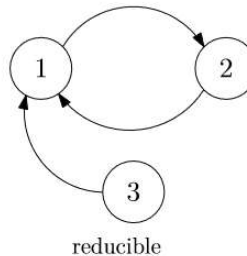
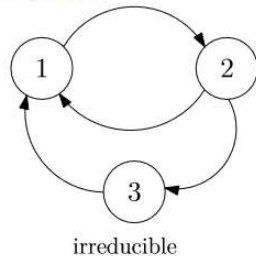


$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots$

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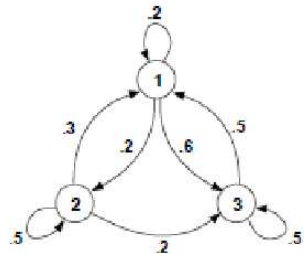
- ▶ Markov chain is **irreducible**: there is a **positive probability** of going from one state to another state in a finite number of steps. Ensures that **every state is visited infinitely often**.



- ▶ Markov chain is **positive recurrent**: starting in any state i , the **expected time to return** to state i is **finite**. Equivalently, the proportion of time spent in every state is positive as length of chain $n \rightarrow \infty$. Ensures that no one state is "left out".

- Example: 3 state chain with transition matrix

$$\mathbf{T} = \begin{bmatrix} 0.2 & 0.2 & 0.6 \\ 0.3 & 0.5 & 0.2 \\ 0.5 & 0.0 & 0.5 \end{bmatrix}$$

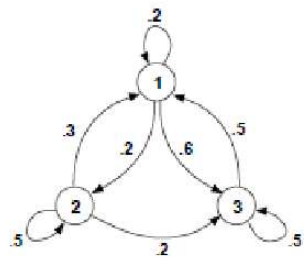


Solve $\pi \mathbf{T} = \pi \Rightarrow \pi = \left[\frac{25}{69}, \frac{10}{69}, \frac{34}{69} \right]$.

$\pi(i) > 0 \quad \sum_i \pi(i) = 1$

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- Fact: a **finite state** irreducible Markov chain is always positive recurrent, therefore it has a stationary distribution.

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$$\begin{array}{ccccccc} & \uparrow & & \uparrow & & & \\ x_0 & \xrightarrow{\quad} & x_1 & \xrightarrow{\quad} & x_2 & \xrightarrow{\quad} & \dots \xrightarrow{\quad} x_k \xrightarrow{\quad} \dots \\ \pi_0 & & \pi_1 = \pi_0 \mathbf{T} & & \pi_2 = \pi_1 \mathbf{T} = \pi_0 \mathbf{T}^2 & & \pi_k = \pi_0 \mathbf{T}^k \end{array}$$
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$$\pi = \lim_{k \rightarrow \infty} \pi_k \text{ is stationary distribution}$$
- If it does, π must be a stationary distribution.

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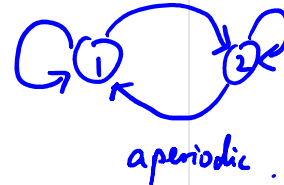
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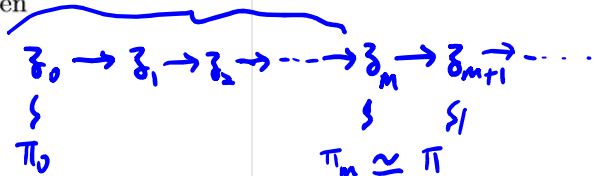
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- Such a MC is said to be **ergodic**.
- We will always assume above conditions hold.
- After a sufficiently long **burn-in** period m , the distribution at each time $k > m$ becomes approximately the stationary distribution.

how to design?
 $\pi \mathbf{T} = \pi$ target



Reversible MC

- *Sufficient* but not necessary condition for π to be stationary is

$$\pi(x)\mathbf{T}(x, y) = \pi(y)\mathbf{T}(y, x)$$

for all x, y . This Markov chain is said to be *reversible*.

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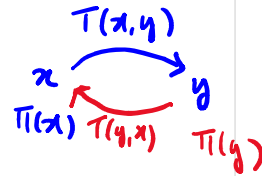
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$$\sum_x \pi(x)\mathbf{T}(x, y) = \sum_x \pi(y)\mathbf{T}(y, x) = \pi(y) \sum_x \mathbf{T}(y, x) = \pi(y) \Rightarrow \pi\mathbf{T} = \pi$$

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 - ▶ all the ergodicity conditions are satisfied
 - ▶ (usually make the Markov chain reversible and aperiodic)
 - ▶ $\pi(x)$ is its stationary distribution.
- We can generate a sample path for the Markov chain $z_0, z_1, \dots, z_n$, starting from **any** z_0 . If n is large, we have $p(z_n) \approx \pi(z_n)$, so we will have obtained a **sample from π !** **This is the basis of MCMC.**

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Metropolis-Hastings Algorithm

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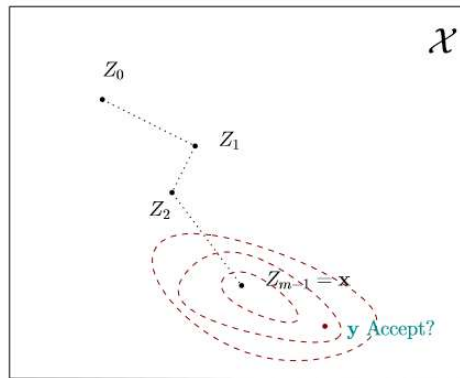
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Let Z_m , $m = 0, 1, \dots$ be the realizations of the following Markov chain.
At step m ,

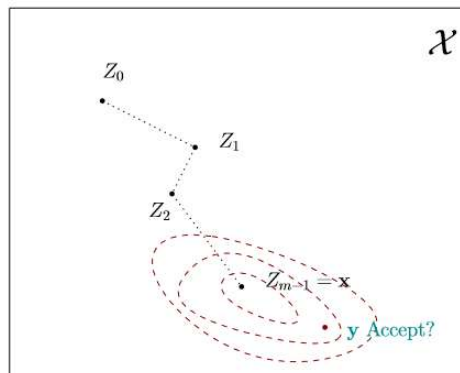


Metropolis-Hastings Algorithm

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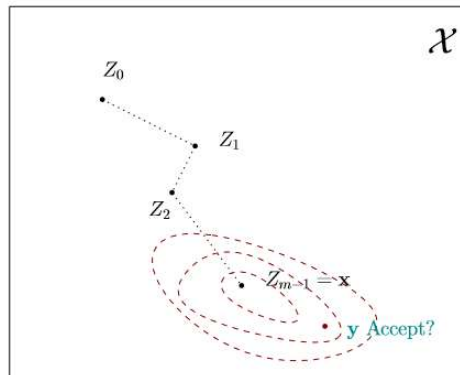
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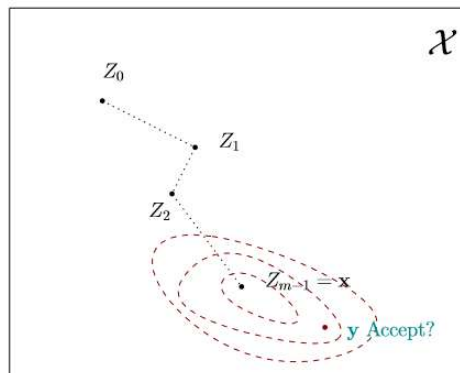
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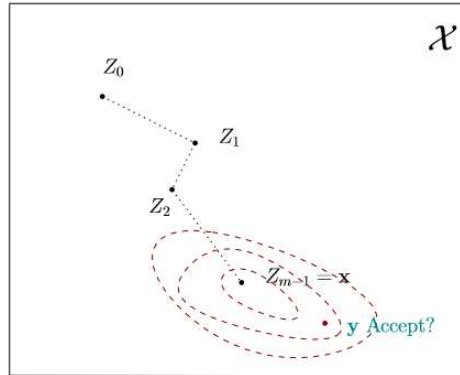
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$$A(\mathbf{x}, \mathbf{y}) = \min \left(1, \frac{\tilde{\pi}(\mathbf{y})q(\mathbf{y}, \mathbf{x})}{\tilde{\pi}(\mathbf{x})q(\mathbf{x}, \mathbf{y})} \right)$$

4. If y is accepted, set $Z_m = y$.
Otherwise, set $Z_m = x = Z_{m-1}$.



z_0 random init.

$\vdots$
 $\downarrow$
 $z_{n-1} =: x$

$$\downarrow \quad q(x, y) = q(y|x)$$

$$A(x, y) = \min(1, \frac{\tilde{\pi}(y) f(y, x)}{\tilde{\pi}(x) g(x, y)})$$

$\neg$ $\rightarrow$ reject w.p. $\pi(x)q(x, y)$
 $\vdash A(x, y)$

Accept
w.p. $A(x,y)$

$$\downarrow$$
$$z_m = z_{m-1}$$
$$Z_m = y$$

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- $Z_0, Z_1, \dots$ is a Markov chain as Z_m depends only on the value of Z_{m-1} .
- Similar to rejection and importance sampling, no need to know normalizing constant for $\tilde{\pi}(\mathbf{x})$.
- Suitable for high dimensional distributions as sampling is from an “easy” proposal $q(\mathbf{x}, \cdot)$.

Proof of MH (Supplementary)

Lemma

The constructed MH Markov chain is reversible and $\pi(\mathbf{x})$ is its stationary distribution.

Proof.

Let $T(\mathbf{x}, \mathbf{y})$ be the transition probability of the MC. We show that the MC is reversible w.r.t. $\pi(\mathbf{x})$. That means $\pi(\mathbf{x})$ is the stationary distribution. For $\mathbf{x} \neq \mathbf{y}$,

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□

- If the MH Markov chain is irreducible, (positive recurrent,) aperiodic $\Rightarrow$ distribution of Z_n approaches $\pi(\cdot)$ as $n \rightarrow \infty$.
- E.g., if state space $\mathcal{X}$ is finite, choose $q(\mathbf{x}, \mathbf{y}) > 0, \forall \mathbf{x}, \mathbf{y} \in \mathcal{X}$.

Proposal Distributions

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- Run the MC for a large number of steps and discard these. This is the burn-in period. Typically, the first 1000 to 5000 samples are discarded.

Standard dist. } $\sim \pi(\cdot)$
 rejection sampling
 importance sampling } $\approx \pi(\cdot)$
 SIR
 MCMC

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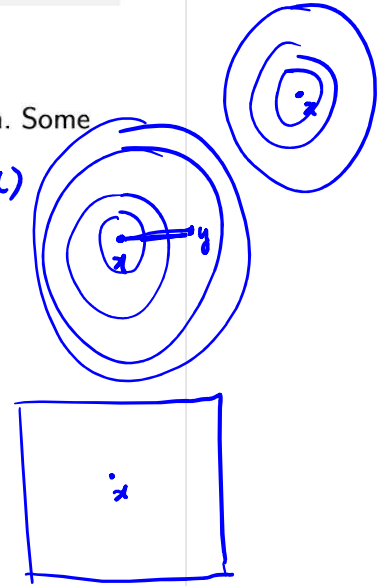
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- $q(x, y) = q(y - x)$ - random walk MH
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 - $y - x \sim \text{Unif}[-\delta, \delta]^d$ - Uniform distribution centered at x

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Remark 1: In these examples, $q(x, y) = q(y, x) \Rightarrow$

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This is known as the Metropolis Algorithm.

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Remark 2: Random walk in the state space. Need to choose the variance appropriately. If variance is small, progress through the state space is slow. If variance is high, we have high rejection rates.

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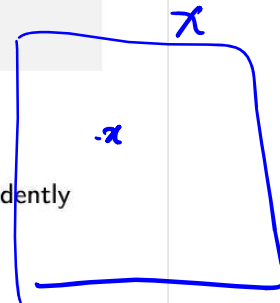
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 - In general, behaves well only if $q(\mathbf{y})$ is more heavy-tailed than $\pi(\mathbf{y})$.
- ▶ **Exploit properties** of $\pi(\mathbf{x})$. Suppose $\pi(\mathbf{x}) \propto \psi(\mathbf{x})h(\mathbf{x})$, where $h(\mathbf{x})$ is a density that can be sampled from, and $\psi(\mathbf{x})$ is a bounded function. Take $q(\mathbf{x}, \mathbf{y}) = h(\mathbf{y})$. Then,

$$\begin{aligned}
 A(\mathbf{x}, \mathbf{y}) &= \min \left(1, \frac{\tilde{\pi}(\mathbf{y})q(\mathbf{y}, \mathbf{x})}{\tilde{\pi}(\mathbf{x})q(\mathbf{x}, \mathbf{y})} \right) \\
 &= \min \left(1, \frac{\psi(\mathbf{y})h(\mathbf{y})h(\mathbf{x})}{\psi(\mathbf{x})h(\mathbf{x})h(\mathbf{y})} \right) \\
 &= \min \left(1, \frac{\psi(\mathbf{y})}{\psi(\mathbf{x})} \right).
 \end{aligned}$$



Wooclap

Course Content



Week 1: Introduction, Probability Review

This week introduces the foundational concepts of pattern recognition and machine learning, emphasizing the importance of probability in these fields.



○ 1_Introduction.pdf



○ **W** Wooclap

Wooclap is an interactive platform which makes courses more interactive and measure the comprehension of students using smartphones.

Example

$\pi(x) \propto \exp(-x^2)x^2(5 + \sin(2x))$. Use a Gaussian transition proposal distribution (random walk MH):

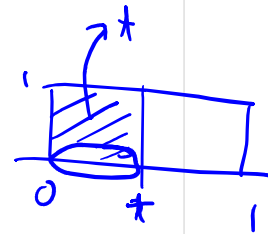
$$q(x, y) = \mathcal{N}(y | x, \sigma^2) = \mathcal{N}(x | y, \sigma^2) = q(y, x)$$

so

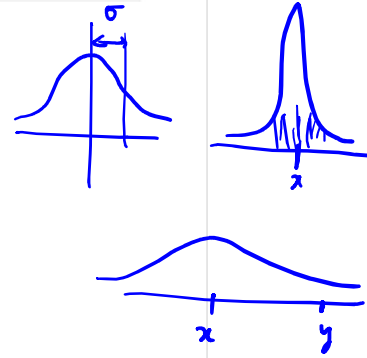
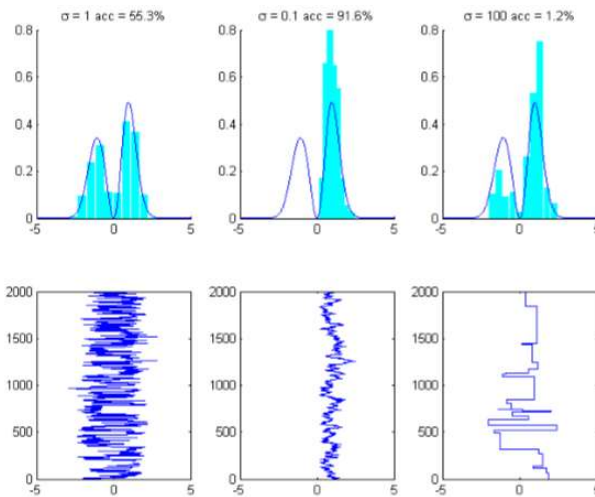
$$\underline{A(x, y)} = \min \left(1, \frac{\tilde{\pi}(y)}{\tilde{\pi}(x)} \right)$$

At each step,

- Sample y from $q(Z_m, y)$.
- Generate uniform random variable $\underline{U} \sim \text{Unif}[0, 1]$. Set $Z_{m+1} = y$ if $\underline{U} \leq A(Z_m, y)$. Otherwise, set $\underline{Z_{m+1}} = Z_m$.



Example



Proposal Variance

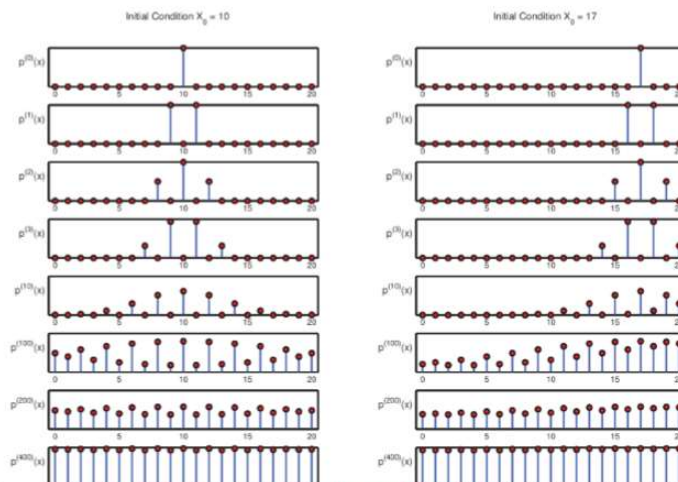
- Important to use appropriate σ .
- σ too small - slow convergence, takes a long time to explore the whole space, acceptance rate too high.
- σ too large - big steps, but very low acceptance rate, gets stuck for a long time.
- Rules of thumb:
 - ▶ random walk MH: acceptance rate of 0.25 to 0.5
 - ▶ independence chain MH: acceptance rate close to 1

Burn-In

- **Burn-in** phase: throw away samples from initial period before chain is close to the target stationary distribution.

Burn-In

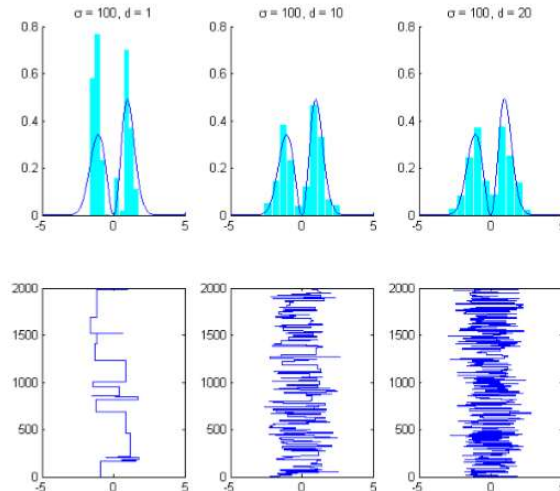
- **Burn-in** phase: throw away samples from initial period before chain is close to the target stationary distribution.
- But **difficult to diagnose** when the chain has burned in. E.g., $\text{Unif}(\{0, 1, \dots, 20\})$. Takes over 400 steps until chain has “forgotten” its starting point.



Thinning

- **Thinning** - break dependence between draws in the MC by taking every d th sample.
- Useful when σ is chosen too large - stuck at same location for long periods of time. Sub-sample every d samples from original MC.

$$x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_m \rightarrow x_{m+1} \rightarrow x_{m+2} \rightarrow \dots$$



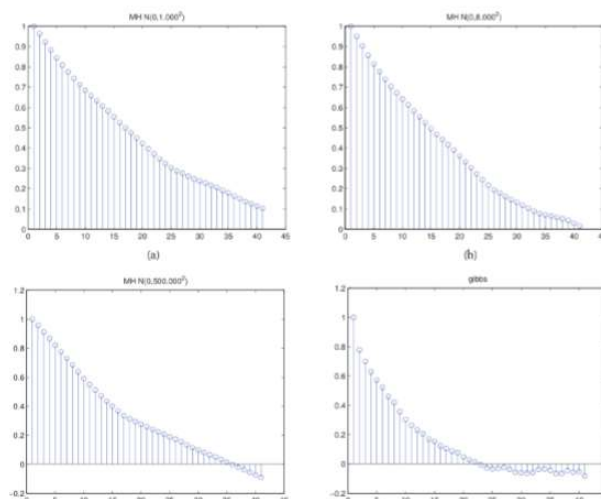
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IE4497/EE6497 Pattern Recognition and Deep Learning

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Python

- [M12]: 06_mcmc_gmm_demo.ipynb
- Use thinning: subsample every d samples.
- Larger $d \Rightarrow$ lower correlation between samples.
- Tradeoff: more samples wasted.



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Outline

- 1 Stationarity
- 2 Metropolis-Hastings
- 3 Gibbs Sampling
- 4 Conclusion
- 5 Some Practice Problems

Gibbs Sampling

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- If we can compute the **full conditionals** $p(z_i | \mathbf{z}_{-i})$, then we can use the Gibbs sampler:

$$\pi(\mathbf{z})$$

$$p(\mathbf{z})$$

↑ ↑

$$p(z_1 | z_2, \dots, z_d)$$

$$p(z_2 | z_1, z_3, \dots, z_d)$$

$$\vdots$$

$$p(z_d | z_1, \dots, z_{d-1})$$

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$$\vec{z} = \begin{bmatrix} z_1 \\ \vdots \\ z_d \end{bmatrix} \rightarrow \text{vector}$$

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▶ Burn-in period samples discarded.

$\uparrow$ scalar $\rightarrow$ one-dimensional

$$z_3^{(k)} \sim p(\cdot | z_1^{(k)}, z_2^{(k)}, z_4^{(k-1)}, \dots, z_d^{(k-1)})$$

Gibbs Sampling: Why it works (Supplementary)

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- Therefore, $\mathbf{y}_j^{(k)}$ has joint pdf $p(z_1, \dots, z_d)$.

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- Equivalent to MH step with transition from $\mathbf{x}_j^{(k)}$ to $\mathbf{y}_j^{(k)}$ given by transition probability $q_j(\mathbf{x}_j^{(k)}, \mathbf{y}_j^{(k)}) = p(z_j^{(k)} | \mathbf{x}_{-j}^{(k)})$ and acceptance probability

$$\begin{aligned}A(\mathbf{x}_j^{(k)}, \mathbf{y}_j^{(k)}) &= \frac{p(\mathbf{y}_j^{(k)})q_j(\mathbf{y}_j^{(k)}, \mathbf{x}_j^{(k)})}{p(\mathbf{x}_j^{(k)})q_j(\mathbf{x}_j^{(k)}, \mathbf{y}_j^{(k)})} = \frac{p(\mathbf{y}_j^{(k)})p(z_j^{(k-1)} | \mathbf{x}_{-j}^{(k)})}{p(\mathbf{x}_j^{(k)})p(z_j^{(k)} | \mathbf{x}_{-j}^{(k)})} \\ &= \frac{p(z_j^{(k)} | \mathbf{x}_{-j}^{(k)})p(\mathbf{x}_{-j}^{(k)})p(z_j^{(k-1)} | \mathbf{x}_{-j}^{(k)})}{p(z_j^{(k-1)} | \mathbf{x}_{-j}^{(k)})p(\mathbf{x}_{-j}^{(k)})p(z_j^{(k)} | \mathbf{x}_{-j}^{(k)})} = 1.\end{aligned}$$

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- ▶ Ignore everything that does not depend on z_1 .
- ▶ Use knowledge of well-known distributions to find the distribution $p(z_1 \mid z_2, \dots, z_d)$.

Example

Gibbs sampling of the following pdf:

$$p(x, y, n) \propto \binom{n}{x} y^{x+a-1} (1-y)^{n-x+b-1} e^{-\lambda} \frac{\lambda^n}{n!}$$

where a, b, λ are positive constants.

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$$p(x) = \iint p(x, y, n) dy dn$$

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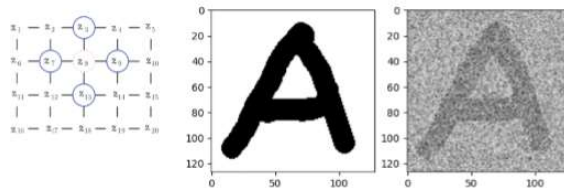
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- This is because it is **hard to find** $p(x)$ directly!



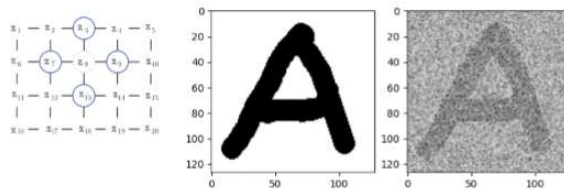
A Real Application: Denoising using the Ising Model

- Image denoising: for a white/black image, each pixel $z_j \in \{-1, 1\}$ is a hidden value of the clean image.



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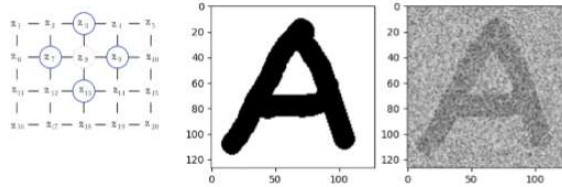
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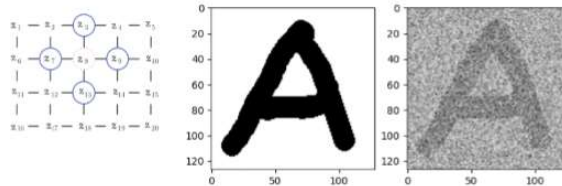
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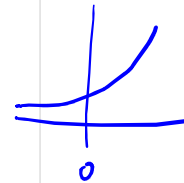
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- Ising prior distribution:

$$p(z_j | \mathbf{z}_{-j}) \propto \prod_{s \in N_j} \psi(z_s, z_j),$$

where N_j is the neighborhood of pixel z_j , and $\psi(u, v) = \exp(Juv)$ with $J > 0$ as the "coupling strength".



$$= \frac{p(y|z)p(z)}{p(y)}$$

$$u, v \in \{+1, -1\}$$

$$e^{Juv} = \begin{cases} e^J & \text{if } u=v \\ e^{-J} & \text{if } u \neq v \end{cases}$$

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Let's compute the Ising prior distribution (we will need it later):

$$\begin{aligned}
 p(z_j \mid \mathbf{z}_{-j}) &= \frac{\prod_{s \in N_j} \psi(z_s, z_j)}{\prod_{s \in N_j} \psi(z_s, z_j) + \prod_{s \in N_j} \psi(z_s, -z_j)} \\
 &= \frac{\exp(J \sum_{s \in N_j} z_s z_j)}{\exp(J \sum_{s \in N_j} z_s z_j) + \exp(-J \sum_{s \in N_j} z_s z_j)} \\
 &= \frac{1}{1 + \exp(-2J \sum_{s \in N_j} z_s z_j)} \\
 &= \text{sigmoid}(2J\eta_j),
 \end{aligned}$$

where $\text{sigmoid}(u) = \frac{1}{1 + \exp(-u)}$ and $\eta_j = \sum_{s \in N_j} z_s z_j$.

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$$\begin{aligned}
 p(z_j | \mathbf{z}_{-j}, \mathbf{y}) &= p(z_j | \mathbf{z}_{-j}, y_j) \quad \text{since } y_j \text{ depends only on } z_j \\
 &= \frac{p(z_j | \mathbf{z}_{-j})p(y_j | z_j)}{p(y_j | \mathbf{z}_{-j})} \quad \text{Bayes' rule} \\
 &= \frac{\text{sigmoid}(2J\eta_j)p(y_j | z_j)}{\text{sigmoid}(2J\eta_j)p(y_j | z_j) + \text{sigmoid}(-2J\eta_j)p(y_j | -z_j)} \\
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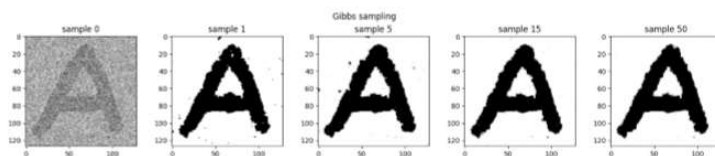
- Reminder: z_j is binary, so it is easy to sample using the above conditional distribution.

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- [M12]: 06_ising_image_denoise_demo.ipynb: interested to find the mean of the posterior distribution $p(\mathbf{z} \mid \mathbf{y})$.

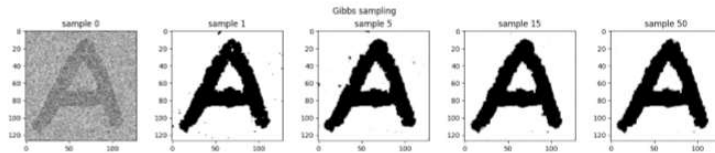
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- Bayesian neural networks: MCMC Training of Bayesian Neural Networks

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- 2 Metropolis-Hastings
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- 4 Conclusion
- 5 Some Practice Problems

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- [06_sampling_summary.ipynb](#)

“... from now on we can compare our data with the model we actually want to use rather than with a model which has some mathematical convenient form. This is surely a revolution.”

Peter Clifford (1993)

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Some Practice Problems

1. Show that in MH sampling, if we were able to sample from $q(x, y) = \pi(y)$, then MH sampling reduces to ordinary sampling.
Solution: At each step, we sample directly from π with acceptance probability

$$A(x, y) = \min\left(1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)}\right) = \min\left(1, \frac{\pi(y)\pi(x)}{\pi(x)\pi(y)}\right) = 1.$$

2. Consider $p(z_1, z_2, z_3) \propto z_1^{z_2+a-2}(1-z_1)^{b-1}\frac{1}{z_2!}\mathbf{1}_{\{z_3 \in [0, z_1]\}}$. Derive all the conditional probabilities used in the Gibbs sampling of (z_1, z_2, z_3) .

Solution:

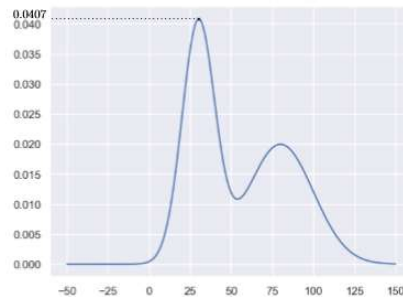
$$p(z_1 | z_2, z_3) \propto z_1^{z_2+a-2}(1-z_1)^{b-1}\mathbf{1}_{\{z_1 \geq z_3\}} \sim \text{Beta}(z_1 | z_2 + a - 1, b) \mathbf{1}_{\{z_1 \geq z_3\}}$$

$$p(z_2 | z_1, z_3) \propto z_1^{z_2}\frac{1}{z_2!} \sim \text{Pois}(z_2 | z_1)$$

$$p(z_3 | z_1, z_2) \propto \mathbf{1}_{\{z_3 \in [0, z_1]\}} \sim \text{Unif}(z_3 | (0, z_1))$$

Some Practice Problems (cont.)

3. Consider the pdf $p(x)$ shown in the following figure.

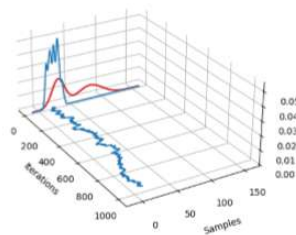


- Julia wishes to sample from the truncated distribution whose pdf is $p_1(x) \propto p(x) \mathbf{1}_{\{0 \leq x \leq 125\}}$. Explain how she can use rejection sampling with the uniform distribution as the proposal distribution.
- Freddy wishes to find the tail probability $\int_{100}^{\infty} p(x) dx$. Explain how this can be achieved using importance sampling.

Some Practice Problems (cont.)

- Russell proposes to use the random walk Metropolis-Hastings algorithm to sample from $p(x)$ using the proposal distribution $q(x, y) = \mathcal{N}(y - x | 0, 1)$. The sampling trajectory is as shown in the following. What is wrong with this approach? What can be done to rectify this?

MH with $N(0, 1^2)$ proposal



Solution:

- Let $\tilde{p}_1(x) = p(x) \mathbf{1}_{\{0 \leq x \leq 125\}}$. The proposal distribution can be $q(x) = \text{Unif}(x | 0, 125)$ and $k = 0.0407/125^{-1} = 5.0875$ so that $kq(x) \geq \tilde{p}_1(x)$ for all x .

Some Practice Problems (cont.)

- b) A proposal distribution like $q(x) = \lambda \exp(\lambda(x - 100)) \mathbf{1}_{\{x > 100\}}$ can be chosen. We draw samples $z_1, z_2, \dots, z_n$ from $q(x)$ and compute $w(z_i) = p(z_i)/q(z_i)$. Finally, we compute the estimate as $\frac{1}{n} \sum_{i=1}^n w(z_i)$.
- c) The standard deviation used in the random walk MH is too small. We can increase it so that the acceptance rate is 0.25 to 0.5. E.g., using s.d. of 15, we have the trajectory in the following.

MH with $N(0, 15^2)$ proposal, acc.=0.26

